Photosynthesis and Chlorophyll Fluorescence: Principles, Regulation, and PAM Fluorometry

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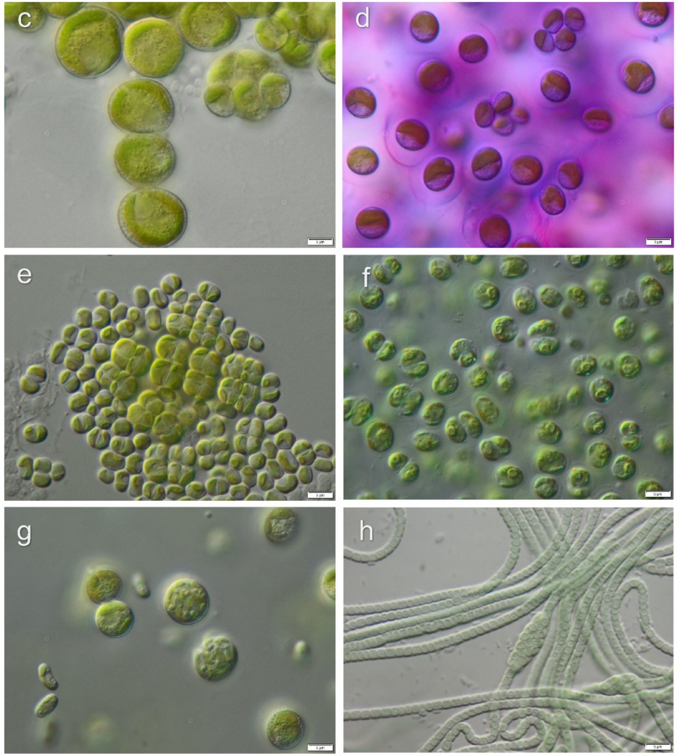
Photosynthetic organisms

Higher plants

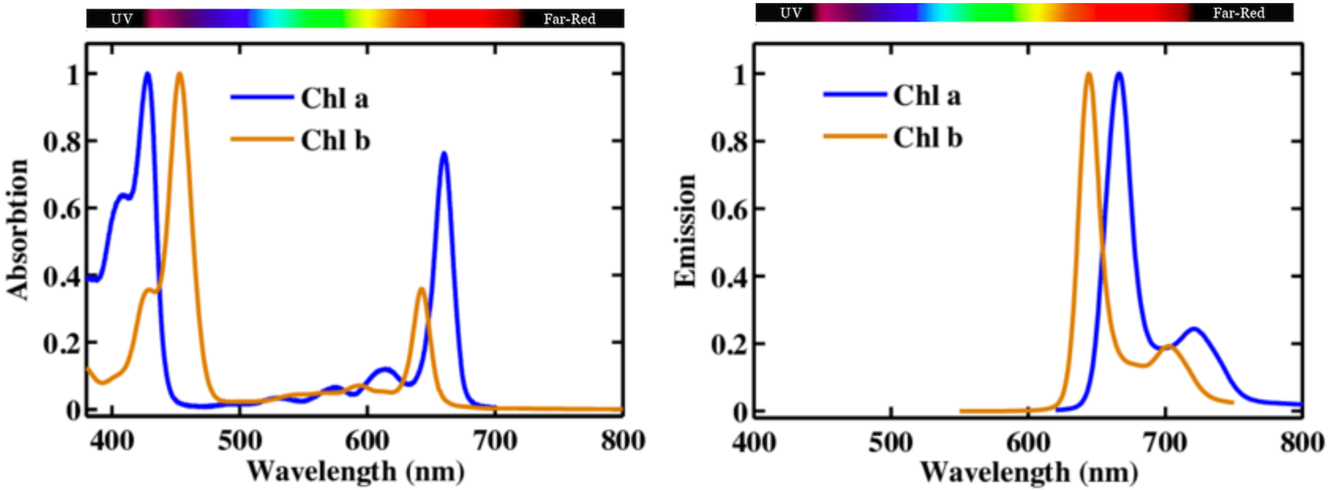
Algae and cyanobacteria



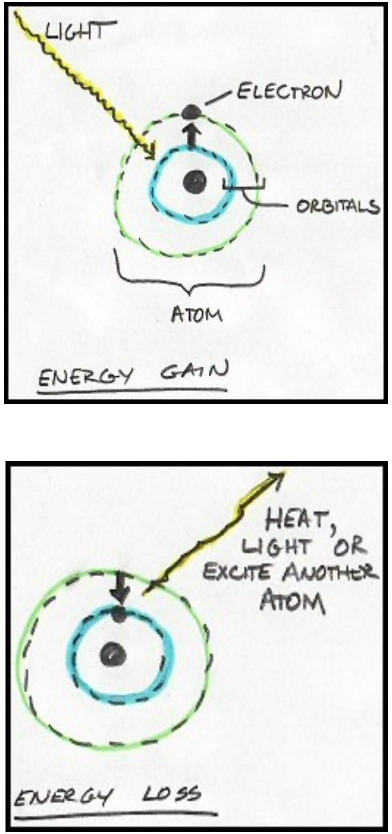
Samolov et al., Microorganisms, 2020

$$\text{water} + CO_2 \xrightarrow{\text{light}} O_2 + \text{sugar}$$

Light reaction

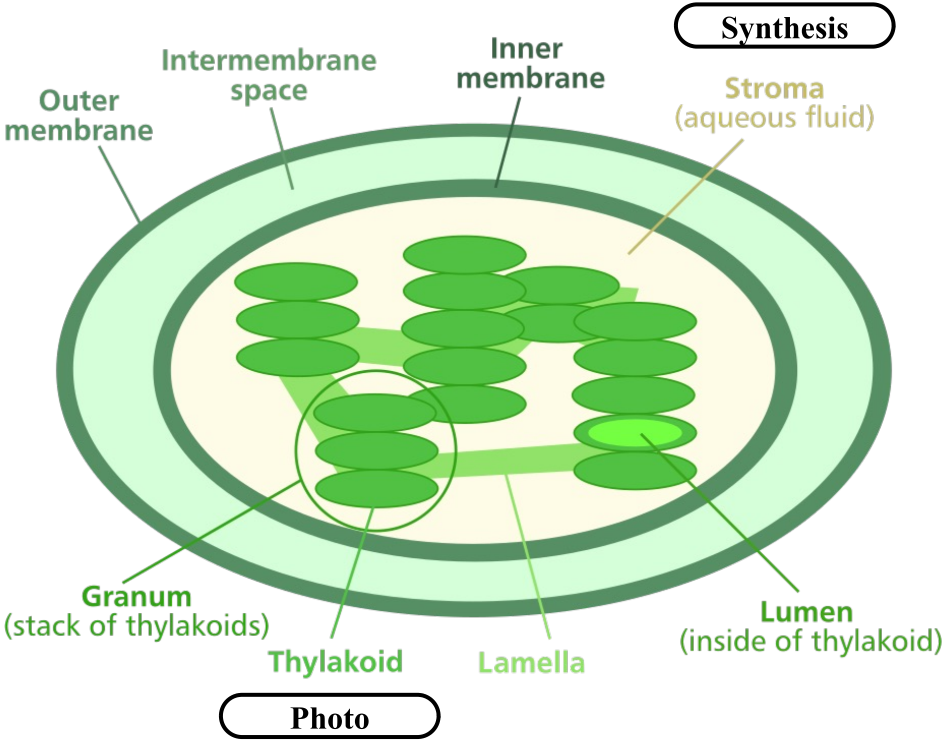


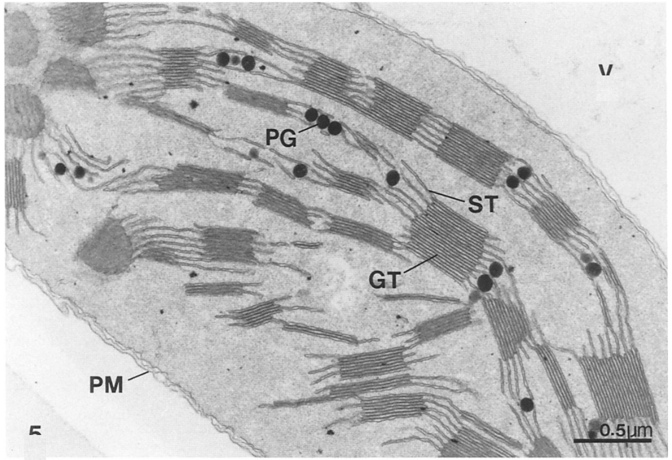
PhD Thesis Mammadov Gulmammad, 2015.



Energy gain and loss. Tom Michaels

Chloroplast





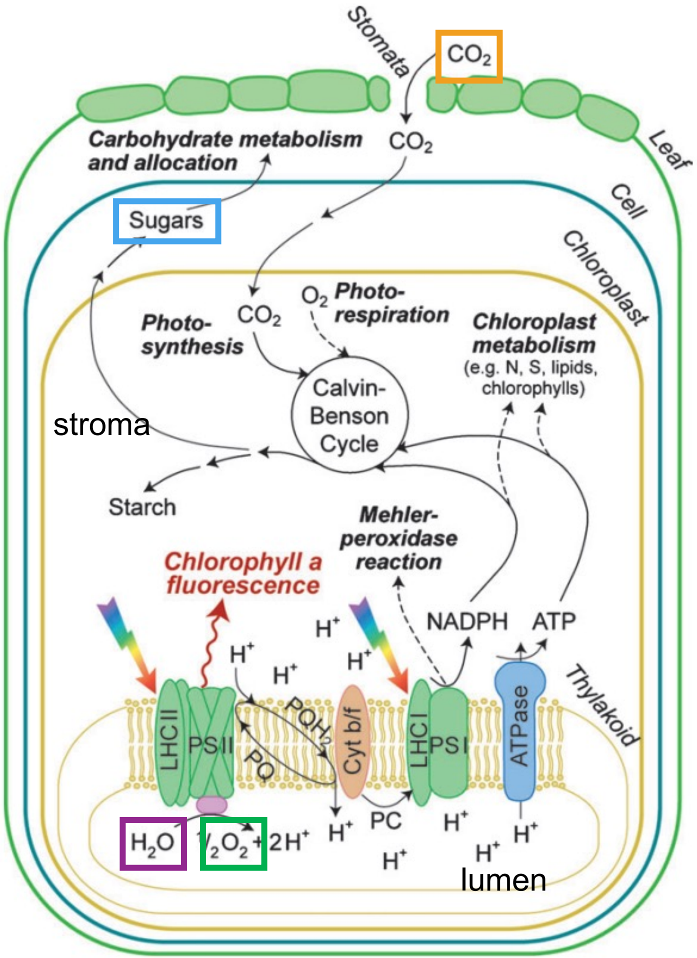
GT: grana thylakoids
ST: stromal thylakoids
PM: plasma Mambrane
PG: plastoglobuli

https://en.wikiversity.org/wiki/Biophysics/Introduction/Chloroplast

Allen et al, The Journal of cell biology, 1988

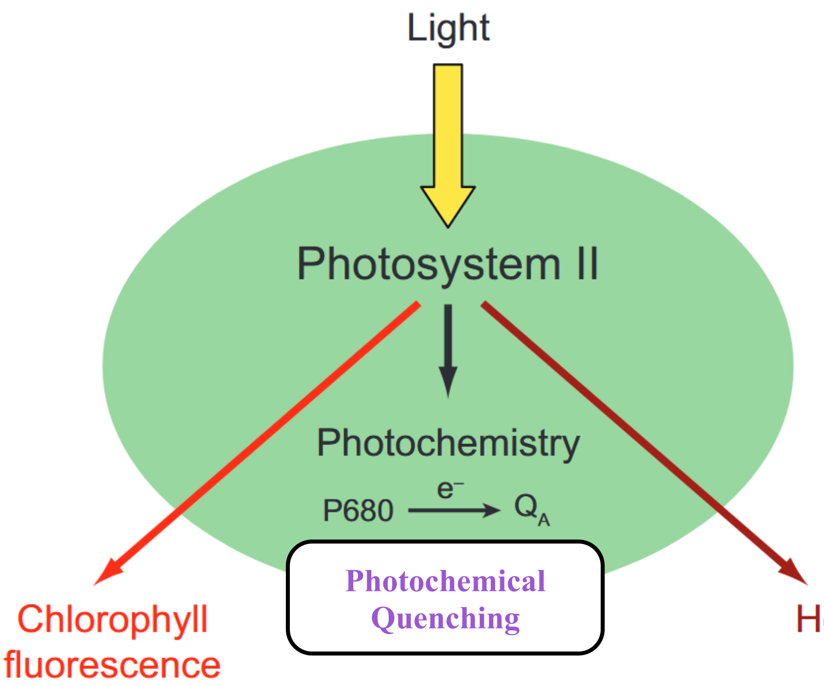
Synthesis

Photo



$$\text{water} + CO_2 \xrightarrow{\text{light}} O_2 + \text{sugar}$$

$[Q_A]$ fraction of oxidised quinone A



Photochemical Quenching

Fluorescence

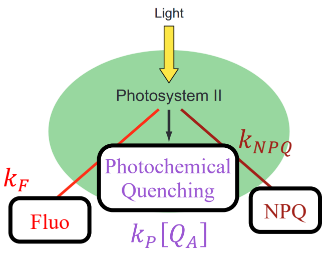
Non-Photochemical Quenching (NPQ)

Fluorescence	Photochemical Quenching	NPQ	
k_F	$k_P[Q_A]$	k_{NPQ}	Dark-adapted
$k'_F = k_F$	$k'_P[Q_A]$	k'_{NPQ}	Light-adapted

Baker et al., Annu. Rev. Plant Biol., 2008

Fluorescence from dark-adapted cells

Dark-adapted

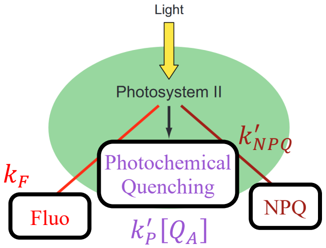


Fluorescence signal proportional to $\Phi_F = \frac{k_F}{k_F + k_P[Q_A] + k_{NPQ}}$

- $F_0 \propto \frac{k_F}{k_F + k_{NPQ} + k_P}$: fluorescence of dark adapted cells ($[Q_A] = 1$).
- $F_M \propto \frac{k_F}{k_F + k_{NPQ}}$: *maximum* fluorescence of dark adapted cells (($[Q_A] = 0$).
- $F_v = F_M - F_0$: variable fluorescence.

Maximum quantum yield of photosynthesis

$$\frac{F_v}{F_M} = \Phi_{PSII}^{\max} = \frac{k_P}{k_F + k_P + k_{NPQ}}$$



Light-adapted

- $F' \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P[Q_A]}$: operating fluorescence in light conditions;
- $F'_0 \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P}$: minimum fluorescence in light conditions;
- $F'_M \propto \frac{k_F}{k_F + k'_{NPQ}}$: maximum fluorescence in light conditions;

$F'_v = F'_M - F'_0$: variable fluorescence in light conditions

$F'_q = F'_M - F'$: quenched fluorescence in light conditions

$$\frac{F'_v}{F'_M} = \Phi_{PSII}^{max}(I) = \frac{k'_P}{k_F + k'_P + k'_{NPQ}}$$

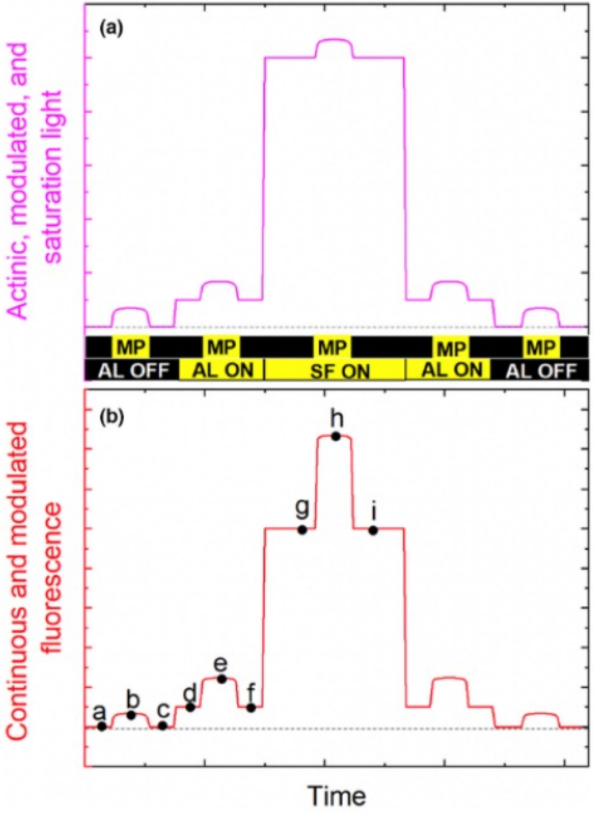
$$\frac{F'_v}{F'_q} = \Phi_{PSII}(I) = \frac{k'_P[Q_A]}{k_F + k'_P[Q_A] + k'_{NPQ}}$$

Absolute fluorescent signal is not required, relative changes are enough.

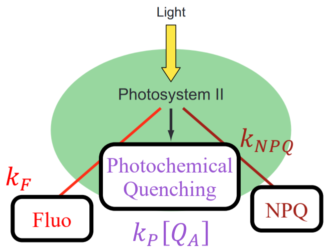
ML (or MP): Mesuring Light (or Pulse), very weak, modulated light used to measure fluorescence without significantly driving photosynthesis

AL: Actinic Light, continuous illumination driving photosynthesis and physiological adaptation.

SP (or SF): Saturating Pulse (or Flash), brief and extremely intense flash that temporarily closes all PSII reaction centres

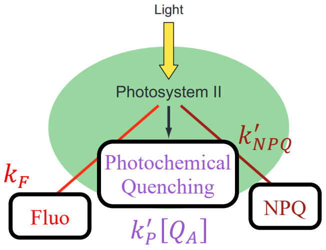


Zuo et al., Plant-Environment Interactions, 2022



$$\frac{F_v}{F_M} = \frac{F_M - F_0}{F_M} = \Phi_{PSII}^{\max}$$

Dark-adapted



$$\frac{F'_v}{F'_M} = \frac{F'_M - F'_0}{F'_M} = \Phi_{PSII}^{\max}(I)$$

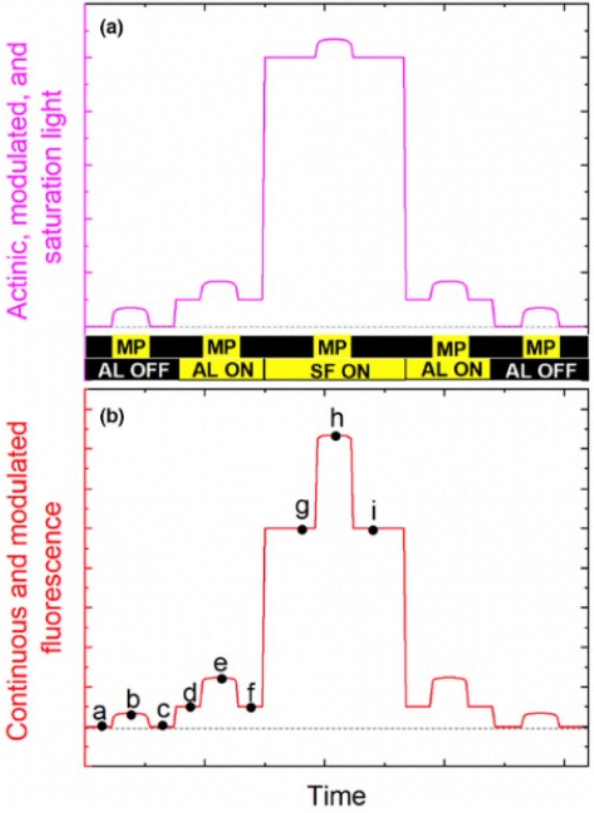
$$\frac{F'_v}{F'_q} = \frac{F'_M - F'_0}{F'_M - F'} = \Phi_{PSII}(I)$$

Light-adapted

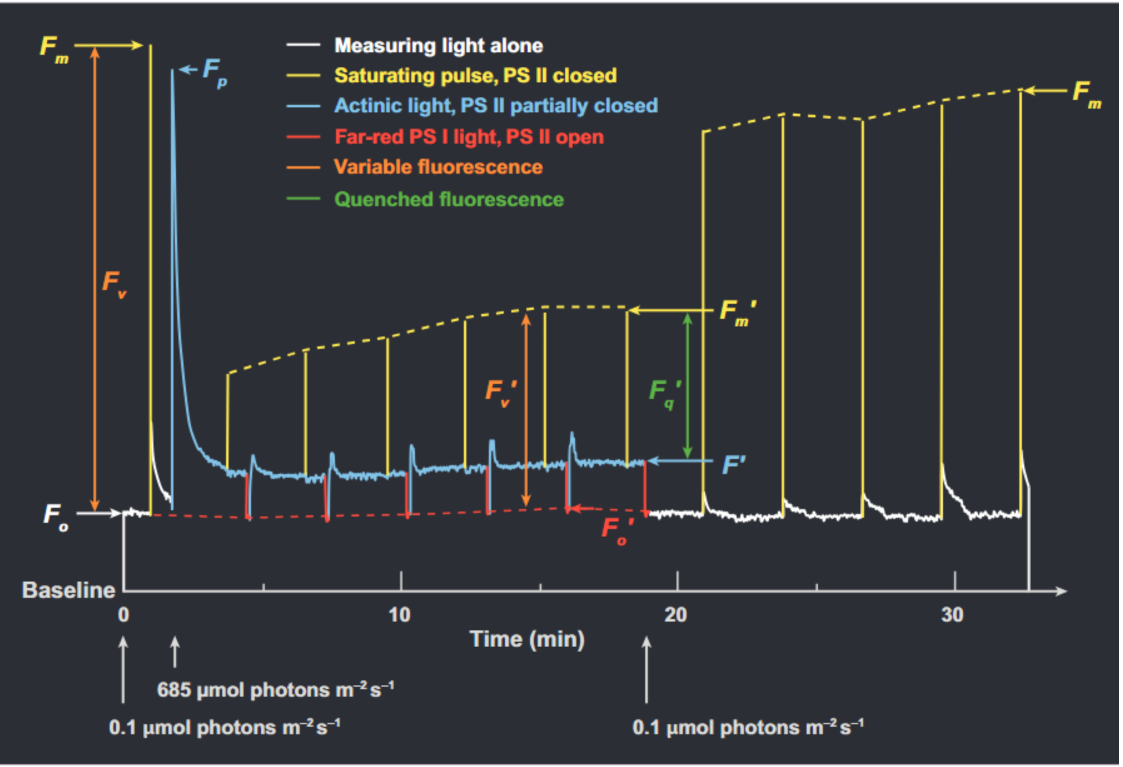
$$F_0 = b - \frac{a + c}{2}$$

$$F'_0 = e - \frac{d + e}{2}$$

$$F'_M = h - \frac{g + i}{2}$$



Zuo et al., Plant-Environment Interactions, 2022



$$\frac{F_v}{F_M} = \frac{F_M - F_0}{F_M} = \Phi_{PSII}^{\max}$$

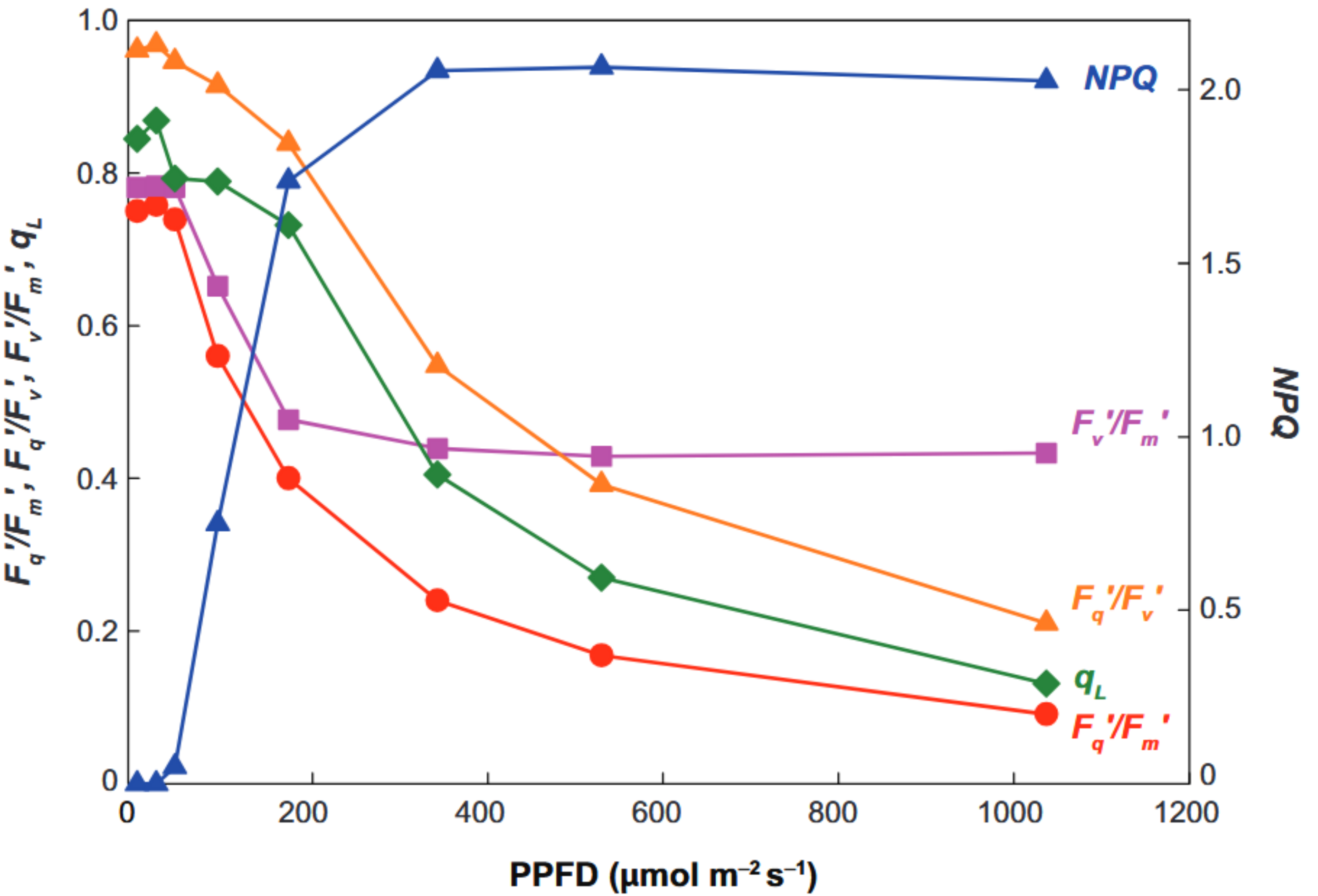
$$\frac{F_v'}{F_M'} = \frac{F_M' - F_0'}{F_M'} = \Phi_{PSII}^{\max}(I)$$

$$\frac{F_q'}{F_M'} = \frac{F_M' - F'}{F_M'} = \Phi_{PSII}(I)$$

$[Q_A]$ can be estimated as $[Q_A] = \left[\frac{F_q'}{F_v'}\right]\left[\frac{F_0'}{F'}\right]$

NPQ can be estimated also as $\frac{F_M - F_M'}{F_M'} = \frac{k_{\text{NPQ}}' - k_{\text{NPQ}}}{k_{\text{NPQ}}}$

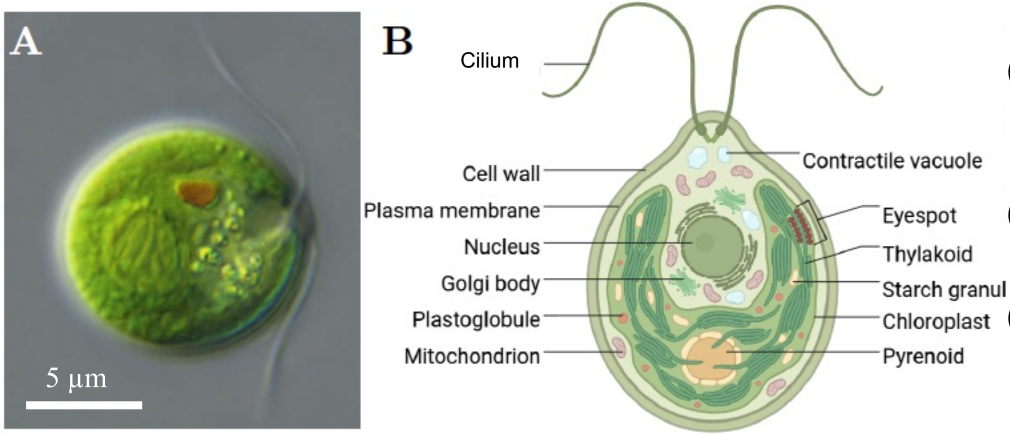
Baker et al., Annu. Rev. Plant Biol., 2008



Baker et al., Annu. Rev. Plant Biol., 2008

Chlamydomonas reinhardtii

[klami] for intimists



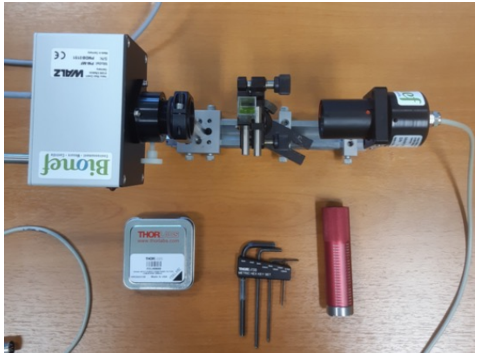
The Science Magazine of the Max Planck Society, 2014

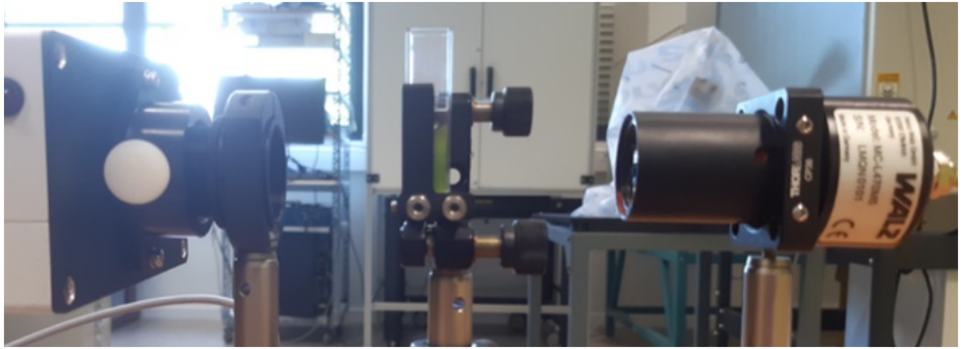
N. Farès, Thèse, 2025

Swimming

Phototaxis

Photosynthesis





ImagingPAM

PhytoPAM ED